

PRINCIPLES OF COMPUTING (4000 BC – 1890)

4000 BC

The abacus was developed and is considered one of the earliest calculating devices in history. It was used by various civilizations to perform basic arithmetic operations and laid the foundation for later calculation systems.

1614

Mathematician John Napier introduced logarithms, a tool that simplified complex mathematical calculations and contributed to the development of future calculating devices.

1642

Blaise Pascal invented the Pascaline, one of the first mechanical calculators. It used gears to perform additions and subtractions automatically.

1671

Gottfried Wilhelm Leibniz developed a more advanced mechanical calculator capable of multiplying, dividing, and performing other mathematical operations, bringing it closer to the concept of a modern calculator.

1801

Joseph Marie Jacquard created an automatic loom controlled by punched cards. This innovation later inspired the programming systems used in the first computers.

1822

Charles Babbage designed the Difference Engine, a machine intended to automate complex mathematical calculations and reduce human errors.

1834

Charles Babbage presented the design of the Analytical Engine, considered the first concept of a general-purpose computer. It included memory, data processing, and programming through punched cards.

1843

Ada Lovelace published notes on the Analytical Engine and included the first algorithm intended to be executed by a machine. For this reason, she is recognized as the world's first computer programmer.

1887–1890

Herman Hollerith developed the punched-card tabulating machine for the 1890 United States Census. His system greatly accelerated data processing and laid the foundation for the future IBM.

Decade of 1930's

1933: Telex messaging network comes online.

Like the Volkswagen Beetle and modern freeway systems, the Telex messaging network comes out of the early period of Germany's Third Reich. Telex starts as a way to distribute military messages, but soon becomes a world-wide network of both official and commercial text messaging that will persist in some countries into the 2000s. Telex uses teleprinters, which date back to the 1910s for use in telegraphy.

But instead of using pricey dedicated telegraph lines, the telex system connects those teleprinters to each other over voice telephone lines, routed by modified telephone switches. Wireless versions of Telex soon connect remote regions of the developing world.

1934: World Brains

Belgian Paul Otlet has a modest goal: collect, organize, and share all the world's knowledge. Otlet had co-created a massive "search engine" starting in the early 1900s. His Mundaneum now combines enhanced card catalogs with sixteen million entries, photos, documents, microfilm, and more. He is working on integrating telegraphy and multiple media, from sound recordings to television. In the 1930s

British writer H.G. Wells and American scientist Vannevar Bush are advancing similar goals—Wells with his "World Brain" writings and Bush with the Memex, a sort of microfilm-based Web browser. These approaches to organizing information differ. But all share key features of today's Web, including automated cross-references – which we call hyperlinks.

1937: Bell Laboratories scientist George Stibitz uses relays for a demonstration adder

Called the “Model K” Adder because he built it on his “Kitchen” table, this simple demonstration circuit provides proof of concept for applying Boolean logic to the design of computers, resulting in construction of the relay-based Model I Complex Calculator in 1939. That same year in Germany, engineer Konrad Zuse built his Z2 computer, also using telephone company relays.

1939: Elektro at the world’s fair

Built by Westinghouse, the relay-based Elektro robot responds to the rhythm of voice commands and delivers wisecracks pre-recorded on 78 rpm records. It appeared at the World's Fair, and it could move its head and arms... and even "smoked" cigarettes.

Hewlett-Packard is founded

David Packard and Bill Hewlett found their company in a Palo Alto, California garage. Their first product, the HP 200A Audio Oscillator, rapidly became a popular piece of test equipment for engineers. Walt Disney Pictures ordered eight of the 200B model to test recording equipment and speaker systems for the 12 specially equipped theatres that showed the movie “Fantasia” in 1940.

Decade of 1940’s

1940: The Complex Number Calculator (CNC) is completed

In 1939, Bell Telephone Laboratories completes this calculator, designed by scientist George Stibitz. In 1940, Stibitz demonstrated the CNC at an American Mathematical Society conference held at Dartmouth College. Stibitz stunned the group by performing calculations remotely on the CNC (located in New York City) using a Teletype terminal connected to New York over special telephone lines. This is likely the first example of remote access computing.

1941: Konrad Zuse finishes the Z3 computer

The Z3, an early computer built by German engineer Konrad Zuse working in complete isolation from developments elsewhere, uses 2,300 relays, performs floating point binary arithmetic, and has a 22-bit word length. The Z3 was used for aerodynamic calculations but was destroyed in a bombing raid on Berlin in late 1943. Zuse later supervised a reconstruction of the Z3 in the 1960s, which is currently on display at the Deutsches Museum in Munich.

The three laws of robotics

Isaac Asimov publishes the science fiction short story *Liar!* in the May issue of *Astounding Science Fiction*. In it, he introduced the Three Laws of Robotics:

A robot may not injure a human being or, through inaction, allow a human being to come to harm.

A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.

A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

This is thought to be the first known use of the term “robotics.”

The first Bombe is completed

Built as an electro-mechanical means of decrypting Nazi ENIGMA-based military communications during World War II, the British Bombe is conceived of by computer pioneer Alan Turing and Harold Keen of the British Tabulating Machine Company.

Hundreds of allied bombes were built in order to determine the daily rotor start positions of Enigma cipher machines, which in turn allowed the Allies to decrypt German messages. The basic idea for bombes came from Polish code-breaker Marian Rejewski's 1938 "Bomba."

1942: The Atanasoff-Berry Computer (ABC) is completed

After successfully demonstrating a proof-of-concept prototype in 1939, Professor John Vincent Atanasoff receives funds to build a full-scale machine at Iowa State College (now University). The machine was designed and built by Atanasoff and graduate student Clifford Berry between 1939 and 1942. The ABC was at the center of a patent dispute related to the invention of the computer, which was resolved in 1973 when it was shown that ENIAC co-designer John Mauchly had seen the ABC shortly after it became functional.

The legal result was a landmark: Atanasoff was declared the originator of several basic computer ideas, but the computer as a concept was declared un-patentable and thus freely open to all. A full-scale working replica of the ABC was completed in 1997, proving that the ABC machine functioned as Atanasoff had claimed. The replica is currently on display at the Computer History Museum.

1943: A logical calculus of the ideas Immanent in nervous activity

Two scientists, Warren S. McCulloch and Walter H. Pitts, publish the groundbreaking paper *A Logical Calculus of the Ideas Immanent in Nervous Activity*. The paper quickly became a foundational work in the study of artificial neural networks and has many applications in artificial intelligence research. In it McCulloch and Pitts

described a simplified neural network architecture for intelligence, and while the neurons they described were greatly simplified compared to biological neurons, the model they proposed was enhanced and improved upon by subsequent generations of researchers.

Bell labs relay interpolator is completed

The US Army asked Bell Laboratories to design a machine to assist in testing its M-9 gun director, a type of analog computer that aims large guns to their targets.

Mathematician George Stibitz recommends using a relay-based calculator for the project. The result was the Relay Interpolator, later called the Bell Labs Model II. The Relay Interpolator used 440 relays, and since it was programmable by paper tape, was used for other applications following the war.

Curt Herzstark designs Curta calculator

Curt Herzstark was an Austrian engineer who worked in his family's manufacturing business until he was arrested by the Nazis in 1943. While imprisoned at Buchenwald concentration camp for the rest of World War II, he refines his pre-war design of a calculator featuring a modified version of Leibniz's "stepped drum" design. After the war, Herzstark's Curta made history as the smallest all-mechanical, four-function calculator ever built.

1944:First colossus operational at Bletchley Park

Designed by British engineer Tommy Flowers, the Colossus is designed to break the complex Lorenz ciphers used by the Nazis during World War II. A total of ten Colossi were delivered, each using as many as 2,500 vacuum tubes. A series of pulleys transported continuous rolls of punched paper tape containing possible solutions to a particular code. Colossus reduced the time to break Lorenz messages from weeks to hours. Most historians believe that the use of Colossus machines significantly shortened the war by providing evidence of enemy intentions and beliefs. The machine's existence was not made public until the 1970s.

Harvard Mark 1 is completed

Conceived by Harvard physics professor Howard Aiken, and designed and built by IBM, the Harvard Mark 1 is a room-sized, relay-based calculator. The machine had a fifty-foot long camshaft running the length of machine that synchronized the machine's thousands of component parts and used 3,500 relays. The Mark 1 produced mathematical tables but was soon superseded by electronic stored-program computers.

1945: John von Neuman writes first draft of a report on the EDVAC

In a widely circulated paper, mathematician John von Neumann outlines the architecture of a stored-program computer, including electronic storage of programming information and data -- which eliminates the need for more clumsy methods of programming such as plugboards, punched cards and paper. Hungarian-born von Neumann demonstrated prodigious expertise in hydrodynamics, ballistics, meteorology, game theory, statistics, and the use of mechanical devices for computation. After the war, he concentrated on the development of Princeton's Institute for Advanced Studies computer.

Plankalkul

Konrad Zuse begins work on Plankalkül (Plan Calculus), the first algorithmic programming language, with the goal of creating the theoretical preconditions for the solution of general problems. Seven years earlier, Zuse had developed and built the world's first binary digital computer, the Z1. He completed the first fully functional program-controlled electromechanical digital computer, the Z3, in 1941. Only the Z4 — the most sophisticated of his creations — survived World War II.

Memex

With side-by-side screens, the imaginary Memex desk is meant to let a user compare and create links between microfilm documents, somewhat like today's clickable Web links and bookmarks. The idea is that people will continually build on each other's associative trails through the world's knowledge, helping tackle the growing problem of information overload. The Memex is the brainchild of top U.S. scientist Vannevar Bush, an analog computing pioneer who had helped oversee development of the atomic bomb. The basic mechanism he suggests is a microfilm automatic selector similar to those built by optics pioneer Emmanuel Goldberg in the early 1930s. Bush publicizes the Memex concept in 1945 articles in *The Atlantic Monthly* and *Life*.

1946: A logic named Joe is published

Under a series of pseudonyms, Will F. Jenkins' science fiction had been regularly appearing since the 1920s. In the March 1946 issue of *Astounding Science Fiction*, Jenkins publishes *A Logic Named Joe*, a short story detailing a world where computers, called Logics, were interconnected into a massive, worldwide network. One such Logic begins to malfunction, giving out secret information about disallowed topics. Almost fifty years prior to the invention of the Internet, *A Logic Named Joe*

contains one of the most prescient views of the capabilities of computers in a network.

Moore School lectures take place

An inspiring summer school on computing at the University of Pennsylvania's Moore School of Electrical Engineering stimulates construction of stored-program computers at universities and research institutions in the US, France, the UK, and Germany. Among the lecturers were early computer designers like John von Neumann, Howard Aiken, J. Presper Eckert and John Mauchly, as well as mathematicians including Derrick Lehmer, George Stibitz, and Douglas Hartree. Students included future computing pioneers such as Maurice Wilkes, Claude Shannon, David Rees, and Jay Forrester. This free, public set of lectures inspired the EDSAC, BINAC, and, later, IAS machine clones like the AVIDAC.

Project Whirlwind begin

During World War II, the US Navy approaches the Massachusetts Institute of Technology (MIT) about building a flight simulator to train bomber crews. Under the leadership of MIT's Gordon Brown and Jay Forrester, the team first built a small analog simulator, but found it inaccurate and inflexible. News of the groundbreaking electronic ENIAC computer that same year inspired the group to change course and attempt a digital solution, whereby flight variables could be rapidly programmed in software. Completed in 1951, Whirlwind remains one of the most important computer projects in the history of computing. Foremost among its developments was Forrester's perfection of magnetic core memory, which became the dominant form of high-speed random access memory for computers until the mid-1970s.

Public unveiling of ENIAC

Started in 1943, the ENIAC computing system was built by John Mauchly and J. Presper Eckert at the Moore School of Electrical Engineering of the University of Pennsylvania. Because of its electronic, as opposed to electromechanical, technology, it is over 1,000 times faster than any previous computer. ENIAC used panel-to-panel wiring and switches for programming, occupied more than 1,000 square feet, used about 18,000 vacuum tubes and weighed 30 tons. It was believed that ENIAC had done more calculation over the ten years it was in operation than all of humanity had until that time.

1947: Manchester Mark I Williams-Kilburn tube

At Manchester University, Freddie Williams and Tom Kilburn develop the Williams-Kilburn tube. The tube, tested in 1947, was the first high-speed, entirely

electronic memory. It used a cathode ray tube (similar to an analog TV picture tube) to store bits as dots on the screen's surface. Each dot lasted a fraction of a second before fading so the information was constantly refreshed. Information was read by a metal pickup plate that would detect a change in electrical charge.

First actual case of bug being found

The word 'bug,' when applied to computers, means some form of error or failure. On September 9th, Grace Hopper records what she jokingly called the first actual computer bug - in this case, a moth stuck between relay contacts of the Harvard Mark II computer prior to its eventual installation at the Naval Weapons Laboratory at Dahlgren. VA.

Hopper helped program the Mark II, and the earlier Harvard Mark I computer, while working for professor Howard Aiken. She worked tirelessly on developing these computers to the fullest through inventive programming. After Harvard, she worked for computer manufacturer Remington-Rand where she developed what is often considered the first compiler, A-0. She also served on the committee to develop COBOL, a standard and widely adopted programming language that transformed the way software was developed for business applications. COBOL is still in use today. Hopper was made a Fellow of the Computer History Museum in 1987.

1948: Cybernetics

Norbert Wiener publishes the book *Cybernetics*, which has a major influence on research into artificial intelligence and control systems. Wiener drew on his World War II experiments with anti-aircraft systems that anticipated the course of enemy planes by interpreting radar images. Wiener coined the term "cybernetics" from the Greek word for "steersman."

The mathematical Theory of communication

American mathematician Claude Shannon writes *The Mathematical Theory of Communication*, laying the groundwork for understanding the theoretical limits of communication between people and machines. As part of this work Shannon identified the bit as a fundamental unit of information and, coincidentally, the basic unit of computation.

First computer program to run on a computer

University of Manchester researchers Frederic Williams, Tom Kilburn, and Geoff Toothill develop the Small-Scale Experimental Machine (SSEM), better known as the Manchester "Baby." The Baby was built to test a new memory technology developed by Williams and Kilburn -- soon known as the Williams Tube -- which was the first high-speed electronic random access memory for computers. Their first program, consisting of seventeen instructions and written by Kilburn, ran on June 21st, 1948. This was the first program in history to run on a digital, electronic, stored-program computer.

SSEC goes on display

The Selective Sequence Electronic Calculator (SSEC) project, led by IBM engineer Wallace Eckert, uses both relays and vacuum tubes to process scientific data at the rate of 50 14 x 14 digit multiplications per second. Before its decommissioning in 1952, the SSEC produced the moon position tables used in early planning of the 1969 Apollo XII moon landing. These tables were later confirmed by using more modern computers for the actual flights. The SSEC was one of the last of the generation of 'super calculators' to be built using electromechanical technology.

1949: Alan Turing quoted by The London Times on artificial intelligence

On June 11, *The London Times* quotes the mathematician Alan Turing. "I do not see why it (the machine) should not enter any one of the fields normally covered by the human intellect, and eventually compete on equal terms. I do not think you even draw the line about sonnets, though the comparison is perhaps a little bit unfair because a sonnet written by a machine will be better appreciated by another machine."

Birth of the Modem

Computers "talk" over ordinary voice phone lines through modems. Developed in 1949 for transmitting radar signals by Jack Harrington's group at the Air Force Cambridge Research Center (AFCRC) near Boston, the modem modulates digital data into sounds, and demodulates received sounds into digital data. (MODulation + DEModulation = MODEM). Modems will be adapted to computers in 1953 for the upcoming SAGE system, and commercialized by Bell Telephone in 1958. By letting computers use normal voice telephone lines, they offer greater coverage and lower costs than dedicated telegraph or leased data lines.

Brain surgeon reflects on artificial intelligence

On June 9, at Manchester University's Lister Oration, British brain surgeon Geoffrey Jefferson states, "Not until a machine can write a sonnet or compose a concerto because of thoughts and emotions felt, and not by the chance fall of symbols, could we agree that machine equals brain – that is, not only write it but know that it had written it. No mechanism could feel (and not merely artificially signal, an easy contrivance) pleasure at its successes, grief when its valves fuse, be warmed by flattery, be made miserable by its mistakes, be charmed by sex, be angry or miserable when it cannot get what it wants."

CSIRAC runs first program

While many early digital computers were based on similar designs, such as the IAS and its copies, others are unique designs, like the CSIRAC. Built in Sydney, Australia by the Council of Scientific and Industrial Research for use in its Radio physics Laboratory in Sydney, CSIRAC was designed by British-born Trevor Pearcey, and used unusual 12-hole paper tape. It was transferred to the Department of Physics at the University of Melbourne in 1955 and remained in service until 1964.

EDSAC completed

The first practical stored-program computer to provide a regular computing service, EDSAC is built at Cambridge University using vacuum tubes and mercury delay lines for memory. The EDSAC project was led by Cambridge professor and director of the Cambridge Computation Laboratory, Maurice Wilkes. Wilkes' ideas grew out of the Moore School lectures he had attended three years earlier. One major advance in programming was Wilkes' use of a library of short programs, called "subroutines," stored on punched paper tapes and used for performing common repetitive calculations within a larger program.

EDSAC

Maurice Wilkes and his team at the University of Cambridge construct the Electronic Delay Storage Automatic Calculator (EDSAC). EDSAC, a stored program computer, used mercury delay line memory. Wilkes had attended the University of Pennsylvania's Moore School of Engineering summer sessions about the ENIAC in 1946 and shortly thereafter began work on the EDSAC.

IBM switches to electronics

IBM executive Thomas Watson Jr., speaking to an IBM sales meeting, predicts that all moving parts in IBM machines would be replaced by electronics within a decade. The change to electronics marked a major shift for IBM, which since the 1920s was the world leader in electro-mechanical punched card systems. Initially, “electronics” meant the use of vacuum tubes for circuitry but in the mid-1950s, IBM management instructed all its engineers to make new equipment: “Solid state by ’58,” was the call--which meant using the new technology of transistors in all new IBM computer products.

MADDIDA developed

MADDIDA is a digital drum-based differential analyzer. This type of computer is useful in performing many of the mathematical equations scientists and engineers encounter in their work. It was originally created for a nuclear missile design project in 1949 by a team led by Fred Steele. It used 53 vacuum tubes and hundreds of germanium diodes, with a magnetic drum for memory. Tracks on the drum did the mathematical integration. MADDIDA was flown across the country for a demonstration to John von Neumann, who was impressed. Northrop was initially reluctant to make MADDIDA a commercial product, but by the end of 1952, six had sold.

MIT - magnetic core memory

While working on the Whirlwind project at MIT, Jay Forrester writes a notebook entry on June 13 that describes his early thoughts on the “coincident current” technique for a magnetic core memory system. This system was the first reliable high-speed random access memory for computers. Magnetic core memory was widely used as the main memory technology for computers well into the 1970s.

Manchester Mark I completed

Built by a team led by engineers Frederick Williams and Tom Kilburn, the Mark I serves as the prototype for Ferranti’s first computer – the Ferranti Mark 1. The Manchester Mark I used more than 1,300 vacuum tubes and occupied an area the size of a medium room. Its “Williams-Kilburn tube” memory system was later adopted by several other early computer systems around the world.

Decade of 1950’s

1950- ERA 1101 Introduced: Designed by ERA and built by Remington-Rand, it was one of the earliest commercially produced computers in history, utilizing a

magnetic drum that stored 1 million bits of data. Its core architectural layout was so successful that it was reused in Remington-Rand systems well into the 1960s.

Magnetic Drum Memory & Atlas: Built for the US Navy to enhance codebreaking capabilities, the "Atlas" stored-program computer was completed, utilizing fixed read/write heads around a rotating ferromagnetic cylinder. Its commercial adaptation was later sold as the ERA 1103.

NPL Pilot ACE Completed: Constructed in Britain based on Alan Turing's philosophy of handling diverse tasks through software programming rather than modifying hardware, this compact machine packed 800 vacuum tubes into just 12 square feet.

SEAC and SWAC Completed: The SEAC was completed in Washington DC as one of the first US stored-program computers, pioneering reliable all-diode logic over vacuum tubes. Concurrently, the SWAC was built at UCLA to solve complex mathematical problems, later executing early climate models

1951- **Ferranti Mark I Sold:** Built as a refinement of the experimental Manchester "Baby," this machine earned the title of the first commercially available general-purpose computer when it was sold to Manchester University. A second unit was later sold to the University of Toronto under the name FERUT.

UNIVAC 1 Delivered: Designed by ENIAC creators J. Presper Eckert and John Mauchly, the massive 29,000-pound UNIVAC 1 became the first commercial computer to capture widespread public attention. Utilizing 5,200 vacuum tubes, Remington Rand sold 46 units at over \$1 million each to corporate, military, and government clients.

LEO-1 Introduced: Modeled directly after Cambridge University's EDSAC, the Lyons Tea Company built the LEO to manage production scheduling and cake delivery logistics. It became England's first commercial computer, performing practical corporate data processing before any other commercial computer system in the world.

1952-**Grace Hopper Completes A-0:** Mathematician Grace Hopper developed A-0 for the UNIVAC I, a pioneering program that allowed users to input instructions using English-like words rather than raw numbers. It possessed several features of a modern-day compiler and transformed how instructions were fed into business computer systems.

IAS Computer Becomes Operational: Created under the supervision of John von Neumann, the Institute of Advanced Study (IAS) computer was fully realized to execute critical calculations for the US atomic weapons program. It solidified the "stored program concept"—storing both data and instructions in shared memory—and its blueprint was copied to create 17 identical systems worldwide, including the MANIAC and ILLIAC.

IBM 726 Magnetic Tape System: IBM announced the Model 726, an essential high-speed magnetic tape system that allowed for inexpensive mass storage by holding 2 million digits per tape. It featured a unique "vacuum channel" method that

allowed the drive to start and stop the circulating tape loop in a split-second, and it was paired commercially with IBM's first electronic digital computer, the Model 701.

1953-Manchester Transistorized Computer Prototype: Richard Grimsdale and Douglas Webb, working under Tom Kilburn at Manchester University, demonstrated the "Manchester TC" prototype. This 48-bit machine utilized 92 point-contact transistors and 550 diodes, marking a vital early step toward transistorized computing.

IBM Ships Model 701: IBM entered the large-scale computer market with the 701 Electronic Data Processing Machine (also known as the "Defense Calculator"), selling 19 units to research labs, aircraft corporations, and the government at a rental price of \$15,000 a month. This milestone marked the beginning of IBM's long-term dominance in the large-scale computer sector.

John Backus Completes Speedcode: John Backus developed Speedcode for the newly introduced IBM 701 scientific computer. While the software consumed a massive amount of the machine's limited memory, it dramatically slashed the time needed to write functional programs, serving as a stepping stone for Backus before he led the historic IBM FORTRAN project.

RAND Corporation Completes Johnniac: Named after computer pioneer John von Neumann, the Johnniac was completed as one of the 17 global systems replicating Princeton's original IAS architecture. Used heavily for complex engineering calculations, it was upgraded for 13 years and hosted pioneering software like the JOSS time-sharing system, which allowed multiple users to access the machine at the same time.

Whirlwind Implements Magnetic Core Memory: MIT's Whirlwind officially became the first computer to deploy functional magnetic core memory. Perfected by Jay Forrester, this highly stable technology arranged tiny magnetic "donuts" onto a wire grid to detect and alter bits, remaining the dominant high-speed RAM standard until the arrival of Intel's DRAM in 1971.

1954-IBM 650 Magnetic Drum Calculator Introduced: IBM launched the Model 650, establishing it as its first mass-produced computer, with 450 units sold in the first year alone. Its magnetic data-storage drum spun at 12,500 rpm, delivering significantly faster data access times than competing drum-based hardware. The system became exceptionally popular across universities, serving as the training ground where an entire generation of students first learned computer programming.

1955- English Electric DEUCE Introduced: This machine was launched as the commercial version of Alan Turing's experimental Pilot ACE, officially named the Digital Electronic Universal Computing Engine. Mostly utilized for complex science and engineering problems alongside a few corporate applications, over 30 units were completed, including one shipped to Australia.

1956- Applying our strict filter to keep only high-level computer hardware, engineering, and architectural milestones for 1956:

- **Direct Keyboard Input to Computers:** Researchers at MIT conducted a highly successful experiment on the MIT Whirlwind computer using an electrically controlled typewriter called a Flexowriter as a keyboard input device. This confirmed the convenience of direct interactive access, laying the groundwork for today's standard mode of operation at a time when users typically fed programs via clumsy punched cards or paper tape.
- **Librascope LGP-30 Introduced:** Defense contractor General Precision purchased physicist Stan Frankel's "MINAC" design and rebranded it as the LGP-30. Selling for less than \$50,000, this single-user system was utilized for scientific, engineering, and basic data processing, standing as an exceptionally early example of a "personal computer".
- **MIT Researchers Build the TX-0:** The TX-0 ("Transistor eXperimental - 0") was completed at MIT's Lincoln Laboratory as the world's first general-purpose programmable computer built using transistors. For easy maintenance, its designers cleverly housed each individual transistor circuit inside a "bottle" that resembled a traditional vacuum tube.
- **IBM Ships RAMAC (First Hard Disk Drive):** The era of magnetic disk storage officially began when IBM shipped the RAMAC 305 computer system to Zellerbach Paper. Utilizing the world's very first hard disk drive, the system featured 50 magnetically coated metal platters storing roughly 5 million characters of data, introducing real-time random access to large amounts of data for the first time.

1957-Digital Equipment Corporation (DEC) Founded: Ken Olsen, Stan Olsen, and Harlan Anderson founded DEC in Maynard, Massachusetts, launching with a \$70,000 venture capital investment from American Research and Development. Initially established to manufacture electronic modules for the test, measurement, prototyping, and control markets, DEC set up its first headquarters inside a leased 19th-century textile mill.

IBM Team Releases FORTRAN: An IBM team led by John Backus developed FORTRAN, a highly powerful scientific computing language that utilized English-like statements. Skepticism about its efficiency compared to manual hand-coding quickly vanished as it proved capable of generating highly efficient machine code, ultimately becoming the most dominant language for scientific and technical computing for decades.

First Computer Scanned Image Completed: Russell Kirsch and his team engineered a rotating drum scanner for the Standards Eastern Automatic Computer (SEAC), enabling the digitization of a five-by-five centimeter black-and-white photograph of Kirsch's son. This breakthrough marked the very first image ever scanned into a computer, laying the operational foundations for modern image processing.

MATH-MATIC and FLOW-MATIC Compilers Released: Sperry Rand rolled out MATH-MATIC, a commercial compiler developed by Grace Hopper for the UNIVAC I as a direct advancement of her earlier A-0 and A-2 compiler concepts. Concurrently, Hopper's foundational compiler research culminated in the completion of B-0 (FLOW-MATIC), which stands as the first English-language compiler designed specifically for corporate business data processing.

1958- RCA Introduces Model 501 Transistorized Computer: Built using a flexible "building block" architecture, the RCA 501 was engineered to scale for diverse corporate needs and could simultaneously control up to 63 magnetic tape drives. This massive data storage capacity made it highly attractive to the US military and large industries, who valued quick database access over raw processing speeds.

SAGE Air Defense System Pioneers Computer Networking: Designed to detect Soviet nuclear bombers, the IBM-built SAGE system pioneered real-time computer networking, interactive computing, and multi-user systems across North America.

The network linked 23 hardened computer centers that communicated simultaneously with radar installations and interceptor aircraft as events unfolded.

SAGE Hardware Infrastructure and Interactive Consoles: The SAGE network deployed massive AN/FSQ-7 computers, each requiring a full megawatt of power to operate 55,000 vacuum tubes, 175,000 diodes, and 13,000 transistors. Hundreds of operators could use the system at the same time, interacting with the airspace display through groundbreaking graphical consoles equipped with an early pointing device known as a light gun.

1959- Bryant Chucking Grinder Company Develops Magnetic Disk Drive: Moving beyond its traditional production of computer drums, the Bryant Chucking Grinder Company engineered a new magnetic disk drive architecture. The hardware consisted of a horizontal shaft outfitted with eight or more massive 39-inch magnesium platters, pushing the early boundaries of physical storage dimensions.

SRI Designs ERMA for Bank of America: SRI International engineered ERMA (Electronic Recording Machine, Accounting), a computer system that completely digitized checking to handle the massive post-WWII surge in check writing. ERMA pioneered a computer-readable font and used a specialized scanner running Magnetic Ink Character Recognition (MICR) to read account numbers preprinted in magnetic ink.

ERMA's Computing Efficiency: The introduction of the ERMA computing system drastically transformed automated data processing speeds. In just one hour of operation, the computer could process a volume of checking accounts that would normally require a well-trained human banker nearly 17 full workdays to complete manually.

Decade of 1960's

1960- COBOL Language Developed: A team from the Pentagon and various manufacturers created COBOL (Common Business-Oriented Language) to achieve machine independence and easy readability. It went on to run the vast majority of business software, accounting for an estimated 80% of all business code by 1997.

DEC PDP-1 Introduced: Priced at \$120,000, this system established the core standards for minicomputers by featuring a CRT graphic display, requiring only one operator, and needing no air conditioning. Its graphic scope allowed early MIT hackers to create *SpaceWar!*, the first computerized video game.

Quicksort Algorithm Created: While studying machine translation in Moscow, C. A. R. Hoare developed Quicksort. It ultimately became one of the most frequently used and foundational sorting methods in computer science history.

1961- CTSS Time-Sharing Demonstrated: Developed at the MIT Computation Center under Fernando Corbató, the Compatible Time-Sharing System (CTSS) was demonstrated on a modified IBM 7090 mainframe. It allowed multiple users to access a single computer simultaneously, pioneering early utilities like text formatting and an inter-user messaging system that presaged modern email.

IBM 1401 Transistorized Mainframe: IBM introduced the 1401, replacing older vacuum tube technology with smaller, more reliable transistors. The system became an enormous success with demand exceeding 12,000 units, resulting in nearly half of all computers in the world being IBM 1401s by the mid-1960s and solidifying the market for general-purpose computers.

UNIMATE Industrial Robot: UNIMATE began working at General Motors, becoming the first mass-produced industrial robot in history. The 4,000-pound robotic arm automated heavy manufacturing tasks by executing step-by-step commands stored directly on a magnetic drum

1962- The Atlas Computer & Virtual Memory: Coming online as the fastest computer in the world, the Atlas introduced the groundbreaking concept of "virtual memory" by using a disk or drum as an extension of main memory. Developed by Tom Kilburn's team at the University of Manchester, this architecture allowed rapid switching between multiple programs, while its system control software, the Atlas Supervisor, became recognized as one of the first true operating systems.

IBM 1311 Removable Disk Drive: IBM announced the 1311, making history as the company's first disk drive to feature a removable disk pack. Each ten-pound pack held 2 million characters, combining the portable convenience of magnetic tapes with the rapid access speeds of hard disks.

MIT LINC Personal Computer: Designed by Wesley Clark at MIT Lincoln Laboratory, the LINC was a pioneering example of a "personal computer" dedicated entirely to a single user. Under an NIH grant, biomedical researchers attended workshops to build their own systems, which were later commercialized by DEC as the LINC-8.

Spacewar! Debuts: Built for the DEC PDP-1 by MIT's Steve "Slug" Russell, Martin "Shag" Graetz, and Wayne Wiitanen, *Spacewar!* debuted as one of the earliest and most influential computerized video games. Its widespread distribution through the DEC users group cemented its legendary status across university and technical computing communities.

1963- **ASCII Standard Introduced:** The American Standard Code for Information Interchange (ASCII) was established, allowing machines from different manufacturers to exchange data using a standardized 128-character set.

Sutherland's Sketchpad: Ivan Sutherland created *Sketchpad*, the first interactive, real-time computer drawing system using a light pen, laying the absolute foundation for modern computer graphics and CAD software.

DAC-1 CAD Program: Developed by General Motors and IBM, DAC-1 became the first commercially available Computer Aided Design program, advancing early time-sharing and display terminal technology.

The Rancho Arm: This six-jointed robotic arm designed for the handicapped was acquired by Stanford University, becoming one of the very first artificial limbs to be controlled by a computer.

Decade of 70's

- **Intel introduces the first microprocessor: 1971**

The first advertisement for a microprocessor, the Intel 4004, appears in Electronic News. Developed for Busicom, a Japanese calculator maker, the 4004 had 2250 transistors and could perform up to 90,000 operations per second in four-bit chunks. Federico Faggin led the design and Ted Hoff led the architecture.

- **The Mark-8 appears in the pages of Radio-Electronics:1974**

The Mark-8 "Do-It-Yourself" kit is designed by graduate student John Titus and uses the Intel 8008 microprocessor. The kit was the cover story of hobbyist magazine Radio-Electronics in July 1974 – six months before the MITS Altair 8800 was in rival Popular Electronics magazine. Plans for the Mark-8 cost \$5 and the blank circuit boards were available for \$50.

- **Apple II introduced:1977**

Sold complete with a main logic board, switching power supply, keyboard, case, manual, game paddles, and cassette tape containing the game Breakout, the Apple-II finds popularity far beyond the hobbyist community which made up Apple's user community until then. When connected to a color television set, the Apple II produced brilliant color graphics for the time. Millions of Apple IIs were sold between 1977 and 1993, making it one of the longest-lived lines of personal computers. Apple gave away thousands of Apple IIs to school, giving a new generation their first access to personal computers.

Decade of 80's

- **IBM introduces its Personal Computer (PC):1981**

IBM's brand recognition, along with a massive marketing campaign, ignites the fast growth of the personal computer market with the announcement of its own personal computer (PC). The first IBM PC, formally known as the IBM Model 5150, was based on a 4.77 MHz Intel 8088 microprocessor and used Microsoft's MS-DOS operating system. The IBM PC revolutionized business computing by becoming the first PC to gain widespread adoption by industry. The IBM PC was widely copied ("cloned") and led to the creation of a vast "ecosystem" of software, peripherals, and other commodities for use with the platform.

- **Sun Microsystems is founded:1982**

When Xerox PARC loaned the Stanford Engineering Department an entire Alto Ethernet network with laser printer, graduate student Andy Bechtolsheim re-designed it into a prototype that he then attached to Stanford's computer network. Sun Microsystems grows out of this prototype. The roots of the company's name came from the acronym for Stanford University Network

(SUN). The company was incorporated by three 26-year-old Stanford alumni: Bechtolsheim, Vinod Khosla and Scott McNealy. The trio soon attracted UC Berkeley UNIX guru Bill Joy, who led software development. Sun helped cement the model of a workstation having an Ethernet interface as well as high-resolution graphics and the UNIX operating system.

- **Intel introduces the 80486 microprocessor:1989**

Intel released the 80486 microprocessor and the i860 RISC/coprocessor chip, each of which contained more than 1 million transistors. The RISC microprocessor had a 32-bit integer arithmetic and logic unit (the part of the CPU that performs operations such as addition and subtraction), a 64-bit floating-point unit, and a clock rate of 33 MHz.

The 486 chips remained similar in structure to their predecessors, the 386 chips. What set the 486 apart was its optimized instruction set, with an on-chip unified instruction and data cache and an optional on-chip floating-point unit. Combined with an enhanced bus interface unit, the microprocessor doubled the performance of the 386 without increasing the clock rate.

Decade of 90's

- **PowerBook series of laptops is introduced:1991**

Apple's Macintosh Portable meets with little success in the marketplace and leads to a complete redesign of Apple's line of portable computers. All three PowerBooks introduced featured a built-in trackball, internal floppy drive, and palm rests, which would eventually become typical of 1990s laptop design. The PowerBook 100 was the entry-level machine, while the PowerBook 140 was more powerful and had a larger memory. The PowerBook 170 was the high-end model, featuring an active matrix display, faster processor, as well as a floating point unit. The PowerBook line of computers was discontinued in 2006.

- **Apple ships the first Newton:1993**

Apple enters the handheld computer market with the Newton. Dubbed a "Personal Digital Assistant" by Apple President John Sculley in 1992, the Newton featured many of the features that would define handheld computers in the following decades. The handwriting recognition software was much maligned for inaccuracy. The Newton line never performed as well as hoped and was discontinued in 1998.

- **IBM releases the ThinkPad 701C:1995**

Officially known as the Track Write, the automatically expanding full-sized keyboard used by the ThinkPad 701 is designed by inventor John Karidis. The keyboard was comprised of three roughly triangular interlocking pieces, which formed a full-sized keyboard when the laptop was opened -- resulting in a keyboard significantly wider than the case. This keyboard design was dubbed "the Butterfly." The need for such a design was lessened as laptop screens grew wider.

Decade of 00's

- **PowerMac G5 is released:2003**

With a distinctive anodized aluminum case, and hailed as the first true 64-bit personal computer, the Apple G5 is the most powerful Macintosh ever released to that point. While larger than the previous G4 towers, the G5 had comparatively limited space for expansion. Virginia Tech used more than a thousand PowerMac G5s to create the System X cluster supercomputer, rated #3 in November of that year on the world's TOP500 fastest computers.

- **Arduino:2005**

Harkening back to the hobbyist era of personal computing in the 1970s, Arduino begins as a project of the Interaction Design Institute, Ivrea, Italy. Each credit card-sized Arduino board consisted of an inexpensive microcontroller and signal connectors which made Arduinos ideal for use in any application connecting to or monitoring the outside world. The Arduino used a Java-based integrated development environment and users could access a library of programs, called "Wiring," that allowed for simplified programming. Arduino soon became the main computer platform of the worldwide "Maker" movement.

- **The MacBook Air is released:2008**

Apple introduces their first ultra notebook – a light, thin laptop with high-capacity battery. The Air incorporated many of the technologies that had been associated with Apple's MacBook line of laptops, including integrated camera, and Wi-Fi capabilities. To reduce its size, the traditional hard drive was replaced with a solid-state disk, the first mass-market computer to do so.

Decade 10's

- **Apple Retina Display:2010**

Since the release of the Macintosh in 1984, Apple has placed emphasis on high-resolution graphics and display technologies. In 2012, Apple introduced the Retina display for the MacBook Pro laptop and iPad tablet. With a screen resolution of up to 400 pixels-per-inch (PPI), Retina displays approached the limit of pixel visibility to the human eye. The display also used In Plane Switching (IPS) technology, which allowed for a wider viewing angle and improved color accuracy. The Retina display became standard on most of the iPad, iPhone, MacBook, and Apple Watch product lines.

- **Raspberry Pi has been developed:2013**

Conceived in the UK by the Raspberry Pi Foundation, this credit card-sized computer features ease of use and simplicity making it highly popular with students and hobbyists. In October 2013, the one millionth Raspberry Pi was shipped. Only one month later, another one million Raspberry Pis were delivered. The Pi weighed only 45 grams and initially sold for only \$25-\$35 U.S. Dollars.

- **The AMD Ryzen Revolution: Saving PC Gaming and Forcing Intel to Adapt: 2017**

By the mid-2010s, the market for desktop and laptop processors (CPUs) was incredibly stagnant. Intel held a near-total monopoly on the market. Because they had no real competition, Intel grew complacent. For years, they released new processor generations that offered measly 5% to 10% performance increases, while keeping high-performance, multi-core computers locked behind premium, incredibly expensive price tags.

Their primary rival, AMD, was on the brink of bankruptcy. Their previous chip architecture (called Bulldozer) was a massive failure, their stock price had plummeted to less than \$2 a share, and tech enthusiasts assumed AMD was effectively dead.

Decade 20's

- **Apple Transitions Away from Intel to Apple Silicon: 2020**

In November 2020, Apple released its first Macs powered by the custom-designed M1 chip, officially beginning its departure from Intel processors. This event shook the personal computer industry by proving that ARM-based architecture—typically reserved for low-power mobile devices—could deliver industry-leading performance and extreme power efficiency in desktop and laptop computers. It forced competitors like Intel, AMD, and Microsoft to aggressively pivot their hardware and operating system roadmaps toward ARM and high-efficiency hybrid architectures.

- **The Generative AI Boom and the Rise of Large Language Models: 2022**

In November 2022, OpenAI released ChatGPT, which triggered an unprecedented paradigm shift in how humans interact with computers. Powered by massive Transformer-based neural networks, this event marked the transition of artificial intelligence from narrow, back-end data processing into a mainstream, accessible interface capable of writing complex code, drafting documentation, and engaging in natural human dialogue. It fundamentally changed software development, cloud infrastructure priorities, and search engine architecture overnight.

- **The Nvidia GPU Data Center Explosion: 2023–2024**

As the demand for AI training and cloud computing skyrocketed, the tech industry experienced a massive hardware shift. Central Processing Units (CPUs), which had traditionally been the core foundation of data centers, took a backseat to Graphics Processing Units (GPUs).

Driven by its proprietary CUDA computing platform and cutting-edge H100 and Blackwell architecture chips, Nvidia's data center revenue grew exponentially, briefly making it the most valuable public company in the world in mid-2024. This event cemented a new era of "accelerated computing," where specialized AI silicon became the most sought-after commodity on Earth, transforming global supply chains and energy grid planning.